

Лабораторная работа №1

Вариант №3

Листинг библиотеки классов (dll):

```
1 using System;
2
3 namespace dll
4 {
5     public class Machine
6     {
7         public string Metal { get; set; } = "None";
8         public int Age { get; set; } = 0;
9         public bool Break { get; set; } = false;
10
11         public void Repair()
12         {
13             if (Break)
14             {
15                 Break = false;
16                 Console.WriteLine("Car is fixed");
17             }
18
19             else
20                 Console.WriteLine("Car is in good condition");
21
22         }
23
24         public void setAge(int new_age)
25         {
26             Age = new_age;
27             Console.WriteLine($"Set age: {Age}");
28         }
29
30         public void ShowInfo()
31         {
32             string Status = Checking();
33             Console.WriteLine($"\\nMachine age: {Age}\\nStatus: {Status}\\nMetal
: {Metal}");
34         }
35
36         public string Checking()
37         {
38             if (!Break)
39                 return "serviceable";
40             else
41                 return "break";
42         }
43     }
44
45     public class Car : Machine
46     {
47         public int id { get; set; } = 0;
48         public string Model { get; set; } = "None";
49         public int Places = 0;
50         public Engine engine = new Engine();
```

```

51     public bool Drive = false;
52
53
54     public void DriveToggle()
55     {
56         if (Drive) { Drive = false; }
57         else { Drive = true; }
58
59         if (Drive && !Break)
60         {
61
62             if (!engine.Work)
63             {
64                 engine.Start();
65             }
66
67             Console.WriteLine("Drive");
68         }
69
70         else if (Break)
71         {
72             Drive = false;
73             Console.WriteLine("Car is broken!");
74         }
75     else
76     {
77         if (engine.Work)
78         {
79             engine.Stop();
80         }
81         Console.WriteLine("Stop drive");
82     }
83 }
84
85
86 public void Honk() { Console.WriteLine("HONK!"); }
87
88 public void ReplaceEngine()
89 {
90     if (engine.Work) { engine.Stop(); }
91
92     Console.WriteLine("Model engine: ");
93     string model = Console.ReadLine();
94
95     Console.WriteLine("Cylinders engine: ");
96     int cylinders = int.Parse(Console.ReadLine());
97
98     Console.WriteLine("HorsePower engine: ");
99     int horse_power = int.Parse(Console.ReadLine());
100
101     Engine new_engine = new Engine { model_engine = model, Cylinders
= cylinders, HorsePower = horse_power };
102     engine = new_engine;
103
104 }
105

```

```

106     public void modelEngine()
107     {
108         Console.WriteLine(engine.model_engine);
109
110     }
111
112     public Car CreateCar()
113     {
114         Car newCar = new Car();
115
116         Console.WriteLine("Model: ");
117         newCar.Model = Console.ReadLine();
118
119         Console.WriteLine("Places: ");
120         newCar.Places = int.Parse(Console.ReadLine());
121
122         Console.WriteLine("--Machine Properties--");
123         Console.WriteLine("Metal: ");
124         newCar.Metal = Console.ReadLine();
125
126         Console.WriteLine("Age: ");
127         newCar.Age = int.Parse(Console.ReadLine());
128
129         Console.WriteLine("Break (true/false): ");
130         newCar.Break = bool.Parse(Console.ReadLine());
131
132         Console.WriteLine("--Engine Properties--");
133         /*newCar.engine = new Engine();*/
134         newCar.engine.createEngine();
135
136         return newCar;
137
138     }
139
140 }
141
142 public class Engine
143 {
144     public int id { get; set; } = 0;
145     public string model_engine { get; set; } = "None";
146     public int Cylinders { get; set; } = 0;
147     public int HorsePower { get; set; } = 0;
148     public bool Work = false;
149
150     public Engine createEngine()
151     {
152
153         Console.WriteLine("Model: ");
154         string Model_engine = Console.ReadLine();
155         Console.WriteLine("Cylinders: ");
156         int cylinders = int.Parse(Console.ReadLine());
157         Console.WriteLine("HorsePower: ");
158         int horsePower = int.Parse(Console.ReadLine());
159
160         return new Engine { Cylinders = Cylinders, HorsePower =
horsePower, model_engine = Model_engine };

```

```

161     }
162
163     public void Start()
164     {
165         if (!Work) { Work = true; } else { Console.WriteLine("The engine
is already running"); }
166     }
167
168     public void Stop()
169     {
170         if (Work) { Work = false; } else { Console.WriteLine("The engine
is no longer running"); }
171     }
172
173
174     public void Upgrade(int extraHorsPower)
175     {
176         HorsePower = extraHorsPower;
177     }
178
179 }
180 }

```

Листинг консольного приложения

```

1 using dll;
2
3 class Program
4 {
5     static public void Main()
6     {
7         bool exit = false;
8
9         do
10        {
11            Console.Clear();
12            Console.WriteLine("1. Create car");
13            Console.WriteLine("2. Info car");
14            Console.WriteLine("0. Exit");
15            int choice = int.Parse(Console.ReadLine());
16
17            switch (choice)
18            {
19                case 1:
20                    Car newCar = new Car();
21                    newCar.CreateCar();
22                    break;
23                case 0:
24                    exit = true;
25                    Console.Clear();
26                    break;
27            }
28
29        } while (!exit);
30    }
31 }

```

Результаты работы программы: